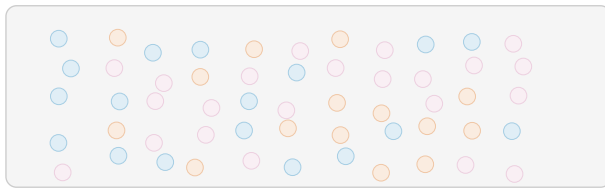


a NOTEARS: Pooled optimization — all samples mixed

Every sample weighted alike: the most dispersed subpopulations dominate the loss

a



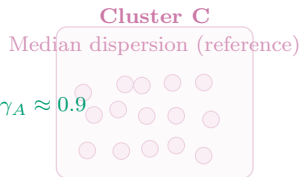
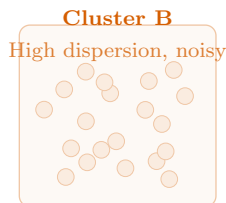
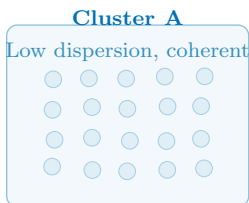
$$\min_W \frac{1}{2n} \|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda_1 \|W\|_1 \quad \text{s.t. } h(W) = 0$$

⇒ **No sample is down-weighted:** the scatter of every subpopulation travels into the structural gradient, injecting spurious edges

b CAGate: Cluster-aware gating — coherent subpopulations weighted, dispersed ones attenuated

Residual-contrast gate γ_k down-weights the clusters whose residuals are the most dispersed

b



$$\gamma_A \approx 0.9$$

$$\gamma_B \approx 0.1$$

$$\gamma_C \approx 0.5$$

$$\min_W \frac{1}{2n} \sum_{k=1}^K \gamma_k \cdot \|\mathbf{X}^{(k)} - \mathbf{X}^{(k)}W\|_F^2 + \lambda_1 \|W\|_1$$

$$\text{s.t. } h(W) = 0, \quad \gamma_k = 1/(1 + e^{-\alpha(\bar{\sigma} - \sigma_k)})$$

⇒ **Precision, not volume:** a smaller graph whose edges are more often confirmed by external databases (29.9% vs 25.4%; STRING ≥ 700)